

TradingAI Current Capabilities and Features

Complete current capability catalog through Milestone 78.

System reference date: 2026-08-29

Standalone current-state reference for the complete platform through Milestone 78.

1. Data acquisition and authority

- Maintain canonical equity/ETF/index universes and governed symbol scope.
- Ingest underlying OHLCV incrementally and backfill multi-year history.
- Ingest Polygon option chains and quotes across governed DTE and liquidity windows.
- Persist historical option observations/features and data-quality diagnostics.
- Ingest earnings events from Alpha Vantage and macro events from Federal Reserve, BLS and BEA sources.
- Ingest futures/OPEX context and pre-open intelligence.
- Assess data freshness, coverage, readiness and publication lineage.

2. Market Overview and market intelligence

- Publish current Market Overview snapshots with bias, health, breadth and regime.
- Measure breadth, participation, trend/momentum, sentiment, sector performance/rotation and leadership.
- Compute correlation/dispersion and cross-asset confirmation/risk context.
- Evaluate dealer positioning, walls/flips, GEX/Vanna/Charm and related flow evidence.
- Integrate volatility/options environment and liquidity/participation into market context.

3. Trend and Stock Intelligence

- Compute daily, weekly and monthly trend states and transitions.
- Forecast trend state and capture institutional participation/leadership evidence.
- Publish current Stock Intelligence for the canonical universe.
- Detect and normalize support/resistance levels, structural zones, breakouts, failures and invalidation levels.
- Rank candidates while preserving underlying thesis, multi-timeframe context, probability/confidence and contradictory evidence.

4. Institutional Options

- Generate strategy alternatives from current Stock Intelligence rather than from a detached option scan.
- Support long options and multi-leg structures including debit/credit verticals, calendars/diagonals and other governed alternatives.
- Select exact Polygon contracts by expiry/strike/right and evaluate liquidity/Greeks.
- Optimize DTE, strike geometry, contract quantity and strategy economics.
- Integrate valuation, event, dealer and market context before handing a candidate to decision/trade-plan layers.

5. Option valuation and relative-value intelligence

- Estimate independent/fair option value and compare with market pricing.
- Analyze implied-volatility surface, skew and term structure.
- Evaluate volatility mispricing, relative value, event mispricing and dealer-flow mispricing.
- Publish explainable valuation-divergence evidence for downstream ranking/decision use.

6. Inflection and timing intelligence

- Compute governed multi-factor and multi-timeframe inflection evidence.
- Use market structure, momentum, participation and related factors to distinguish thesis quality from entry timing.
- Feed timing evidence into Stock/Decision/Options intelligence without allowing a single factor to override governance.

7. Institutional Decision Intelligence

- Combine probability, distribution/tail risk, valuation, inflection, market context, contradictions and governance.
- Produce accept/reject/warning explanations with explicit supporting and contradictory reasons.
- Require current authority lineage rather than silently consuming stale/legacy decisions.
- Feed portfolio-aware decisions into Trade Builder and execution.

8. Portfolio risk and capital allocation

- Aggregate portfolio Greeks, beta-weighted exposure, concentration, correlation and tail/stress risk.
- Evaluate candidate portfolio fit, diversification and opportunity cost.
- Optimize a feasible candidate set under capital, heat, symbol, sector, strategy and correlation constraints.
- Publish current portfolio allocation and recommended actions.
- Rank best-next-trade candidates in the context of the existing portfolio.

9. Advanced Trade Builder

- Convert an approved opportunity into a concrete multi-leg trade plan.
- Preserve exact contract identity, quantity, debit/credit, max loss/profit, reward/risk and net Greeks.
- Define entry, stop, targets, structural invalidation and management logic.
- Revalidate strategy-aware and portfolio-aware conditions before execution.
- Enforce expiration exit rules, including closure before the earliest leg expiration for multi-leg structures.

10. Execution and working-order management

- Stage IBKR paper orders using canonical contract/order identity.
- Normalize prices to IBKR market-rule/minimum-tick requirements.
- Perform freshness/adverse-drift/executable-price preflight.
- Manage working-order age, bounded chasing/repricing, hard timeouts and terminal cancellation.
- Reconcile broker truth and support controlled retry/rearm after terminal cancellation when policy permits.
- Support multi-leg BAG/strategy execution where broker validation allows.

11. Broker truth and portfolio synchronization

- Synchronize IBKR account, cash, positions, orders and fills into PostgreSQL.
- Treat broker truth as authoritative for actual fill/cancel/position state.
- Reconcile local order/position lifecycle after disconnects, cancellation ambiguity or restart.
- Maintain portfolio/account publications used by downstream risk and management services.

12. Autonomous position management

- Continuously monitor filled positions and strategy-level multi-leg state.
- Evaluate stop, target, structural, time, risk and expiration conditions.
- Submit governed exits and reconcile broker fills/cancellations.
- Expose autonomous versus manual management visibility and operational health.

13. Performance learning

- Compare predictions with realized outcomes.
- Evaluate probability calibration and attribution.
- Separate thesis, valuation, execution and management contributions to P&L.
- Analyze execution quality, management quality and trade lifecycle outcomes.
- Provide governed evidence for future research/change proposals.

14. Production operations and live governance

- Run scheduled/continuous services through launchd and application orchestration.
- Monitor health, freshness, dependencies, replay/recovery and alerts.
- Support environment isolation, approvals, audit, rollback/disaster-recovery controls and kill-switch concepts.
- Fail closed when required authorities are stale, missing, incompatible or uncertified.

15. Historical and prospective research - M77

- Replay historical underlying-driven intelligence in isolated research namespaces.
- Run long-history experiments without changing production behavior.
- Freeze prospective candidate cohorts/protocols and accumulate matured outcomes.
- Maintain certified production decisions/rejections separately from challenger research.
- Keep M77 historical discovery closed while prospective certification programs accumulate evidence.

16. Setup Intelligence and conditional alpha - M78

- Detect first-class trend pullback/continuation, breakout/breakdown lifecycle, failed breakout/breakdown reversal, support/resistance reversal and PEAD research setups.
- Persist immutable point-in-time setup snapshots and idempotent lifecycle transitions.
- Attach matured M77 barrier outcomes to setup evidence.
- Assess setup-specific readiness and train hierarchical empirical challengers only when gates pass.
- Explicitly approve and activate shadow models; no automatic activation.
- Generate setup probabilities, expected return/R, uncertainty, capital/time efficiency, shadow ranking and option-expression guidance.
- Accumulate frozen prospective evidence and create certification records without production-promotion authority.

17. User workstation capabilities

- Navigate Market Overview, scanners, institutional intelligence, options valuation, portfolio intelligence, Trade Builder, Execution Workspace and performance/learning views.
- Inspect explanations, warnings, lineage, freshness and governance state instead of receiving opaque trade signals.
- Drill from a ranked underlying candidate through exact option structures, portfolio fit, trade plan and execution.

18. Current operational schedule

Service	Time	Days	Purpose
M71 futures pre-open	07:55	Mon-Fri	Pre-open futures intelligence.
M69.6 event intelligence	08:10	Mon-Fri	Daily earnings/macro event intelligence.
M69.6 morning ingestion	08:30	Mon-Fri	Underlying/market intelligence refresh.
M69.6 intraday options	09:30-14:30 hourly	Mon-Fri	Underlying-coupled intraday options intelligence.
M69.6 EOD ingestion	15:20	Mon-Fri	End-of-day refresh/finalization.
Combined M77 -> M78 shadow	18:30	Mon-Fri	Prospective research; M78 runs only after M77 READY.

19. Important current limitations / intentional boundaries

- M78 currently has no production authority and no active shadow model at the observed evidence level.
- Research success does not automatically alter production weights, thresholds, strategy selection, portfolio allocation, execution or management.
- Paper trading/broker integration is the active governed execution environment described by this documentation; live-capital authority requires separate certification/governance.
- An available feature does not override freshness, lineage, risk or certification gates.